



Have trend-following signals in commodity futures markets become less reliable in recent years?

Benjamin R. Auer^{1,2,3} 

Accepted: 24 February 2021 / Published online: 15 April 2021
© Swiss Society for Financial Market Research 2021

Abstract

Various trend-following trading rules have been shown to be valuable for predicting market directions and thus the formulation of investment strategies. However, recent equity market research has provided striking evidence that the predictive power of such rules appears to diminish over time due to increased investor attention and lowered arbitrage barriers. Given that trend-following rules are also very successful and have been widely used in futures markets, we analyze whether a similar effect can be observed for commodity futures contracts. Using a trend regression approach based on time-varying success ratios, we detect significantly higher predictive accuracy for cross-sectional than for time-series strategies. In addition, with the exception of a few commodities, we find no significant trending behavior in trading rule reliability. These results, which are robust in a variety of settings, indicate strong momentum stability in futures markets and justify the application of this class of trading rules in commodity futures investing.

Keywords Trend-following rules · Moving averages · Momentum · Trends · Commodity futures

JEL Classification G10 · G17 · C10

✉ Benjamin R. Auer
auer@b-tu.de

¹ Chair of Finance, Brandenburg University of Technology Cottbus-Senftenberg,
Erich-Weinert-Str. 1, 03046 Cottbus, Germany

² Department of Finance, University of Leipzig, Grimmaische Str. 12, 04109 Leipzig, Germany

³ Research Network Area Macro, Money and International Finance, CESifo Munich, Schackstr. 4,
80539 Munich, Germany

1 Introduction

Trend-following trading rules are strategies where one bets that past asset performance will continue into the near future. Such strategies can be implemented in the cross-sectional and the time-series dimensions (see Szakmary et al. 2010). Cross-sectional momentum involves taking long positions in assets that performed relatively well (past winners) and short positions in assets that performed relatively poorly (past losers). This approach, dating back to Jegadeesh and Titman (1993, 2001), has been shown to earn significant abnormal returns in a wide variety of markets (see Swinkels 2004; Fama and French 2012) and is often considered to be the most important capital market effect discovered so far (see Fama and French 2008). Time-series momentum can be motivated by the technical trading rules of Brock et al. (1992), where the crossing of moving averages indicates whether a long (short) position should be entered to benefit from an initiated upward (downward) trend.¹ Rather than looking at the relative returns of several assets in the cross section, moving averages focus on the history of a single asset and also contain valuable investment information (see Kavajecz and Odders-White 2004; Fifield et al. 2008).²

Unfortunately, studies on trend-following strategies in equity markets are often criticized on the grounds that their profits may be illusory (see Foltice and Langer 2015). For example, Korajczyk and Sadka (2004) argue that, after direct and indirect transaction costs, cross-sectional momentum strategies would not have yielded abnormal returns prior to the decimalization of stock price quotes. These difficulties are exacerbated by Lesmond et al. (2004) showing that momentum returns are mainly generated by return continuation among poorly performing stocks. Selling such stocks short, as required by a momentum strategy, could prove difficult due to the uptick rule, the inability to gain access to short sale proceeds and the general difficulty associated with borrowing shares of small, illiquid stocks. Similar issues have been raised by Bajgrowicz and Scaillet (2012) and Taylor (2014), highlighting that the performance of typical trend-based technical trading rules is offset by the introduction of low transaction costs, crucially relies on the ability of investors to short-sell stocks and appears to be confined to particular periods.

Partly as a result of these frictions, attention has recently shifted to the examination of trend-following strategies in commodity futures markets, where, as discussed by Shen et al. (2007) and Marshall et al. (2008), nearby contracts for standard commodities are typically very liquid, cheap to trade, easier to short and allow more realistic portfolio sizes than typical academic momentum stock portfolios containing hundreds or thousands of stocks.³ Shen et al. (2007) and Miffre and Rallis (2007) show that cross-sectional momentum strategies in commodity futures earn impressive returns that are too large to be subsumed by the low transaction costs prevailing in these mar-

¹ Moskowitz et al. (2012) alternatively predict time-series continuation and reversal via regressions of (a) scaled returns on past scaled returns and (b) scaled returns on past return signs.

² Time-series momentum is a timing strategy, whereas cross-sectional momentum is a selection strategy.

³ Furthermore, futures contracts (and futures-based momentum portfolios) are excellent diversifiers and inflation hedges, especially when they are actively managed (see Gorton and Rouwenhorst 2006; Erb and Harvey 2006; Miffre and Rallis 2007; Miffre and Fernandez-Perez 2015).

kets.⁴ In the latter study from 1979 to 2004, tactical allocation in commodity futures markets has generated significantly positive annual returns of about 10%, whereas an equally weighted portfolio of commodity futures loses about 3% per year. Szakmary et al. (2010) and Clare et al. (2014) provide similar evidence for various parameterizations of dual moving average (crossover and channel) strategies after data mining adjustments and transaction costs. Finally, the results of Fuertes et al. (2010, 2015), de Groot et al. (2014) and Bianchi et al. (2015a) highlight that trend-following strategies can even be enhanced by combining them with other selection variables like term structure signals, reversal measures or idiosyncratic volatilities.⁵

While this performance is quite impressive, recent stock market research doubts that the predictive power of trend-following trading rules can be considered consistent over time. Several authors have provided striking evidence that many variables which have been shown to be good predictors of future stock returns tend to lose this capability over time. For example, Chordia et al. (2014) highlight that the recent regime of reduced transaction costs, increased trading activity and thus higher market liquidity is associated with an attenuation of a dozen prominent equity return strategies (including momentum and reversal) due to increased arbitrage. Similarly, McLean and Pontiff (2016) and Jacobs and Müller (2020) show that the out-of-sample and post-publication predictive power of about 100 traditional forecasting variables (again including momentum) is significantly lower than the in-sample results presented in earlier studies. They conclude that this finding is related to data mining effects and the fact that investors tend to adjust their trading behavior based on what they learn from academic publications. As summarized by Park and Irwin (2007) and supported by Strobel and Auer (2018), with similar explanations of data mining or adaptive markets, the predictive power of technical trading rules in equity markets also appears to weaken over time.

This brings us directly to the question of whether such tendencies also arise in commodity futures markets. So far, there is only limited evidence on potential time-varying predictive power of trend-following trading rules in these markets, even though mutual fund investments and computerized trade have been continuously increasing here as well (see Irwin and Yoshimaru 1999; Hong and Yogo 2012). Some subsample robustness checks performed in previous studies suggest that the forecasting accuracy may change over time. For example, Park and Irwin (2005) and Szakmary et al. (2010) show that the returns of various trend-following strategies are lower in the most recent subperiods of their studies. Furthermore, Narayan et al. (2015) highlight lower returns in the period covering the financial crisis of 2007. Unfortunately, such examples do not allow the deduction of general tendencies. This is why we step into the picture and systematically study the predictive power of trend-following trading rules in commodity futures markets over time.

⁴ Transaction costs in futures markets range from 0.0004 to 0.033% (see Fleming et al. 1996; Locke and Venkatesch 1997), which is much less than the conservative 0.5% estimate of Jegadeesh and Titman (1993) or the more realistic 2.3% estimate of Lesmond et al. (2004) for the equity market.

⁵ For an excellent review of additional techniques to strengthen traditional momentum signals, see Miffre (2016). Also note that Fuertes et al. (2015) introduce a very intuitive method of forming bivariate and trivariate portfolios (via combined ranking scores) which might set an important standard for future research.

Within a sample of 24 commodity futures indices covering almost 50 years of data, we implement several specifications of cross-sectional and time-series momentum rules, which previous studies have labeled profitable, and document their trading signals. In a first contribution, we evaluate their predictive power by calculating a popular metric of forecasting accuracy: the success ratio h , which gives the percentage of correct signals (see Morana 2001; Wang et al. 2004; Katusiime et al. 2015). This ratio is important because previous studies do not report the signal reliability behind the monetary value of the strategies. Our more detailed analysis can reveal whether the strategies often foresee relative commodity positions and market direction or their returns are mainly generated by only a few correct signals. In a second contribution, we investigate the evolution of predictive power over time by implementing a trend regression approach similar to Olson (2004), Chordia et al. (2014) and Strobel and Auer (2018). If we found downward trends in predictive power, this could be interpreted as evidence of adaptive markets where investors react to the previous success of trading rules such that—amplified by the additional trend of increasing market liquidity and algorithmic trading activity (see Hendershott et al. 2011; Hong and Yogo 2012)—it is now more difficult to predict based on these rules than it was in the past. In contrast, insignificant trending behavior or significantly positive trends would indicate that future performance is likely to be similar to the past or even higher, respectively.

The remainder of our article is organized as follows. Section 2 offers a brief description of our data set. Section 3 discusses our momentum specifications and our trend regression setup. Section 4 contains our main results and the outcomes of some robustness checks. Section 5 concludes and outlines directions for future research.

2 Data

For our analysis from January 1970 to December 2019, we use a selection of commodity futures which guarantees high liquidity and investability. Specifically, we follow Erb and Harvey (2006), Bianchi et al. (2015a, b, 2016) and Zaremba (2016) by concentrating on the 24 commodities listed in the Standard & Poors Goldman Sachs Commodity Index (S&P GSCI): six energy products (Brent crude oil, WTI crude oil, gas oil, heating oil, natural gas and unleaded gasoline), two precious metals (gold and silver), five industrial metals (aluminum, copper, lead, nickel and zinc), eight agricultural products (cocoa, coffee, corn, cotton, soybeans, sugar, Chicago wheat and Kansas wheat) and three livestock products (feeder cattle, lean hogs and live cattle). For each commodity, there exists a daily total return index which represents an investment in fully collateralized nearby futures and thus can be considered a useful representation of commodity returns actually available to investors. We obtain the index data from Thomson Reuters Datastream.⁶

In each index, futures contracts are rolled forward at the beginning of their expiration months, always keeping the investment in nearby futures. Table 1 illustrates that many commodities (e.g., in the energy and industrial metals sectors) have liquid futures

⁶ Because the properties of the corresponding returns have been extensively documented (see, for example, Auer 2015; Zhang et al. 2018), we omit descriptive statistics illustrating non-normality and serial correlation. However, they are available upon request.

contracts that expire every month and are thus rolled forward every month. In contrast, other commodities (e.g., agricultural and livestock products) have only a few contract months each year that trade with sufficient liquidity. These commodities, with futures that expire less frequently, are rolled forward less frequently than every month. As specified in S&P Dow Jones Indices (2016), an expiring futures contract is rolled over to a contract further from expiration at a rate of 20% per day for the five days of the roll period (i.e., from the fifth to the ninth business day of a month).⁷ Until just before the end of the fifth business day, we are invested in the expiring contract. At the end of the fifth day, the holdings are adjusted so that 20% new contracts are held and 80% remain in the expiring contract. The roll process continues on the sixth, seventh and eighth business days with relative weights of the old to the new contract of 60%/40%, 40%/60% and 20%/80%. At the end of the ninth business day, the roll is completed with a position of 100% in the new contract. This kind of adjustment is similar to the approach of Kho (1996) which eliminates any price gap at each joint point of two successive contracts. Otherwise time-series momentum rules could generate spurious signals. Furthermore, in contrast to an immediate rollover, the smoothing of the rollover ensures lower impact of big financial investors on prices when they sell before expiration.

Finally, note that, even though not all commodity futures have data over the entire sample period, our analysis can be performed without obstacles. This is because, consistent with actual investor behavior, we include only the commodities with available data when periodically deriving the commodity rankings for the cross-sectional strategies. Similarly, the time-series strategies for the individual commodities simply start when there is sufficient data for their implementation.

3 Methodology

3.1 Trend-following rules

Cross-sectional momentum Cross-sectional momentum strategies are typically implemented by buying assets with strong past performance and selling assets with weak past performance, where cumulative returns have become the standard measure for past performance (see Yao 2012). When specifying a momentum strategy, it is important to adequately set the ‘evaluation period’ (the time period in which past performance is measured) and the ‘holding period’ (the time period in which long and short positions are held until a new evaluation of winners and losers is performed). While, in equity markets, the evaluation period typically covers the past 12 months without the most recent month—because a short-term reversal effect (past winners (losers) tend to become future losers (winners)) exists when using only the performance of the most recent month—and the holding period is usually one month (see Chordia et al. 2014), settings in futures markets are slightly different. Here, researchers typically use evaluation periods covering the most recent 1, 2, 3, 6, 9 or 12 months (see Bianchi

⁷ In contrast, Dow Jones UBS commodity indices have a slightly different rollover, from the sixth to the tenth business day (see Bianchi et al. 2015a).

Table 1 Commodity futures contracts

Commodity	Trading facility	Ticker	Expiration months	Availability
Energy				
Crude oil (Brent)	ICE UK	LCO	All	1999
Crude oil (WTI)	NYM/ICE	CL	All	1987
Gas oil	ICE UK	LGO	All	1999
Heating oil	NYM	HO	All	1983
Natural gas	NYM/ICE	NG	All	1994
Unleaded gasoline	NYM	RB	All	1988
Precious metals				
Gold	CMX	GC	Feb, Apr, Jun, Aug, Dec	1977
Silver	CMX	SI	Mar, May, Jul, Sep, Dec	1973
Industrial metals				
Aluminum	LME	MAL	All	1991
Copper	LME	MCU	All	1977
Lead	LME	MPB	All	1995
Nickel	LME	MNI	All	1993
Zinc	LME	MZN	All	1991
Agriculture				
Cocoa	ICE US	CC	Mar, May, Jul, Sep, Dec	1984
Coffee	ICE US	KC	Mar, May, Jul, Sep, Dec	1981
Corn	CBT	C	Mar, May, Jul, Sep, Dec	1970
Cotton	ICE US	CT	Mar, May, Jul, Dec	1977
Soybeans	CBT	S	Jan, Mar, May, Jul, Nov	1970
Sugar	ICE US	SB	Mar, May, Jul, Oct	1973
Wheat (Chicago)	CBT	W	Mar, May, Jul, Sep, Dec	1970
Wheat (Kansas)	KBT	KW	Mar, May, Jul, Sep, Dec	1999
Livestock				
Feeder cattle	CME	FC	Jan, Mar, Apr, May, Aug, Sep, Oct, Nov	2002
Lean hogs	CME	LH	Feb, Apr, Jun, Jul, Aug, Oct, Dec	1976
Live cattle	CME	LC	Feb, Apr, Jun, Aug, Oct, Dec	1970

For the 24 commodities covered by the S&P GSCI (i.e., the corresponding futures contracts), this table shows the trading facility, the ticker, the expiration months and the starting year of data availability in Thomson Reuters Datastream. Trading facility abbreviations are used as follows: CBT = Chicago Board of Trade, CME = Chicago Mercantile Exchange, CMX = New York Commodities Exchange, ICE UK (US) = Intercontinental Exchange United Kingdom (United States), KBT = Kansas City Board of Trade, LME = London Metal Exchange, NYM = New York Mercantile Exchange. The term ‘all’ refers to all calendar months from January to December

et al. 2015a) and often focus exclusively on 1-month holding periods (see Szakmary et al. 2010). This is because the short-term reversal effect documented in stock markets does not occur in futures markets (see Miffre and Rallis 2007) or is limited to specific research settings (see Wang and Yu 2004; Lubnau and Todorova 2015). Furthermore, the highest returns of momentum strategies are typically observed for a 12-month

evaluation period and 1-month holding period (see Erb and Harvey 2006; Bianchi et al. 2015a).⁸

We derive buy and sell signals similar to Fong et al. (2005). That is, each day we rank all eligible commodities independently on the basis of past total returns, where the return for each commodity during the evaluation period is measured as the total log percentage change in the index value. Taking into account the previous evidence, we use five evaluation periods: 1, 3, 6, 9 and 12 months. As in Szakmary et al. (2010), we then assume that a 1-month long position is taken in a commodity if it is a past winner (top third of commodities based on the evaluation period return), a 1-month short position is entered if it is a past loser (bottom third) and no position is held if it ranks in the middle. This way we can document daily trading signals (long, short, neutral) for each commodity.⁹ Note that we rank daily and not only at the end of the month for two reasons. First, to obtain meaningful annual success ratios, we require more than twelve signals a year. Second, our approach allows a more general analysis of predictive power because it also covers the perspective of investors trading on days other than the end of the month.

Time-series momentum Time-series momentum can be implemented via a variable moving average (VMA) strategy. According to this strategy, a long position is taken if an asset's short-term moving price average S_t (over s days) exceeds its long-term moving price average L_t (over l days) by w percent, i.e., if $S_t > L_t(1 + w)$, and a short position is taken if $S_t < L_t(1 - w)$. No position is entered when S_t is within the specified moving average band, i.e., when $L_t(1 - w) \leq S_t \leq L_t(1 + w)$. To avoid the pernicious effects of data mining and to follow typical settings used in investment practice, previous equity market studies have often concentrated on five (s, l, w) rules (see Brock et al. 1992; Taylor 2000): (1, 50, 0.01), (1, 150, 0.01), (5, 150, 0.01), (1, 200, 0.01) and (2, 200, 0.01), where s and l are given in trading days. Typical continuous rule application can lead to frequent position changes at a day trader level (see Park and Irwin 2007).

Interestingly, while considerable research has been directed to the study of technical rules in equity markets only a handful of studies have considered futures markets (see Marshall et al. 2008; Clare et al. 2014). While most of these studies use in-sample optimization to derive optimal values for s , l and w , recent work is guided by practitioners' choices because Park and Irwin (2010) have shown that data snooping biases can have crucial effects when trading in futures markets. Furthermore, we can observe that different data frequencies are used to calculate moving averages. For example, Szakmary et al. (2010) use monthly data and thus have specifications of (1, 6, 0.025), (1, 12, 0.05), (2, 6, 0.025) and (2, 12, 0.05) where s and l are given in months.¹⁰

⁸ Miffre and Rallis (2007) show that profitability declines with rising holding period length. They even document negative (zero) average returns over horizons of 18–24 months (beyond 24 months).

⁹ Similar to Szakmary et al. (2010), we do not form one momentum portfolio based on all commodities but base our analysis on individual securities. Our cross-sectional ranking simply serves the purpose of generating winner/loser signals which may be compared with the actual winner/loser position in the holding period.

¹⁰ The value of w is different from typical stock market settings because this way the strategies generate neutral signals in about one-third of the time and are thus comparable to the cross-sectional strategy outlined above.

In contrast, Narayan et al. (2015) analyze both monthly and daily frequencies. The short-run moving averages range from 1 to 12 months (10 to 40 days) and the long-run moving averages range from 2 to 36 months (50 to 200 days).¹¹ They consider end-of-month rule updating (and thus 1-month holding periods) and show that both frequencies lead to successful strategies.¹²

In our implementation, we follow Marshall et al. (2008) by using daily data in moving averages. Translating Szakmary et al. (2010), our daily rule specifications are (20, 120, 0.025), (20, 240, 0.05), (40, 120, 0.025) and (40, 240, 0.05). Similar to our cross-sectional strategies and for comparability of results, we generate signals on a daily basis and assume 1-month holding periods.

3.2 Success ratio regression

To evaluate the predictive power of a strategy for a specific commodity, we compare its forecasts to the actual realizations. In the case of cross-sectional momentum, we check whether predicted winners/losers are in fact winners/losers in the holding period succeeding a signal. Similarly, for time-series momentum, we consider the predicted market direction (buy signal = upward, sell signal = downward) and the actual market movements. Because we focus on a 1-month holding period, we derive actual realizations by looking at the relative magnitudes and signs of the (cumulative) returns in the 20 trading days after a signal, respectively.

By performing this comparison, we can identify the numbers of correct (helpful) and incorrect (misleading) signals. Relating the former to their sum gives a strategy success ratio h (see Katusiime et al. 2015). If we calculate the success ratio each year, we obtain a time-series of annual success ratios $\{h_\tau\}_{\tau=1}^m$, where τ is an index number representing the years in the sample of a given commodity. Put differently, h_1 , h_2 and h_m are the success ratios for the first, second and last sample year, respectively.

To judge the existence of potential trends in the success ratios, we follow Strobel and Auer (2018) by regressing h_τ on time.¹³ That is, we perform the least squares regression $h_\tau = \alpha + \beta \cdot \tau + \epsilon_\tau$, where α and β are the regression constant and slope, respectively, τ is the explanatory variable and ϵ_τ is a classic error term. In this setting, a statistically significant $\beta < 0$ ($\beta > 0$) indicates a downward (upward) trend in the success ratio and thus decreasing (increasing) predictive success of a given strategy over time. α can be interpreted as an initial level of success, which is lowered ($\beta < 0$) or raised ($\beta > 0$) when moving forward in time.

¹¹ Note that Narayan et al. (2015) calculate moving averages based on returns instead of prices.

¹² In a related area of research, Narayan et al. (2013) find that commodity futures are better predictors for spot markets when daily data are used instead of monthly data.

¹³ Strobel and Auer (2018) emphasize that alternatively conducting trend regressions based on strategy returns leads to very similar trend conclusions because success ratios and strategy returns are naturally linked.

4 Empirical analysis

4.1 Main results

We start discussing our results by presenting the time-series averages (and corresponding t-test significances) of the annual success ratios in Table 2. In general, a success ratio of 0.5 indicates that the forecasting effectiveness of a given rule (with respect to relative rank or direction) is identical to tossing a fair coin. In other words, it has a 50:50 chance of generating a correct signal. Thus, if the success ratios of a strategy are on average significantly above (below) 0.5, it is better (worse) than tossing a coin.

As far as cross-sectional momentum is concerned, Table 2 shows that, with few exceptions, success ratios tend to be significantly above 0.5 on average. In line with previous evidence on the potential of such strategies in commodity markets, we often observe the highest success ratios when using a 12-month formation period to capture past performance. For three energy commodities (Brent crude oil, gas oil and unleaded gasoline), the mean success ratios are not significantly different from 0.5 over all formation periods.¹⁴ For lead, three formation periods (1, 6 and 9 months) show insignificance. Mean success ratios below 0.5 (albeit not in a statistically significant way) only occur for the aforementioned energy commodities.

The picture for time-series momentum strategies is quite different. Here, more averages are below 0.5 and fewer are above 0.5. Furthermore, in the latter case, the number of significant outcomes is noticeably lower. Beating a simple coin toss is possible for selected specifications in natural gas, aluminum, nickel, zinc, coffee and sugar markets. In the cases of coffee and sugar futures, all strategy specifications work well. In contrast, for gold, cocoa, soybeans, feeder cattle and lean hogs, tossing a coin tends to have a higher rate of directional effectiveness.

Tables 3 and 4 report the constant and slope coefficient estimates arising in our regressions of annual success ratios on time, respectively. For the cross-sectional momentum strategies, we can make two important observations. First, except for a slightly weaker extent in the energy sector, we have high and significant initial levels of predictive success across all commodities.¹⁵ In a ranking of commodity futures, the relative positions of gold and zinc could have been forecasted best. Second, even though downward trends outnumber upward trends, only few of the former and almost none of the latter are statistically significant. Consequently, success ratios do not deviate strongly from their initial levels over time. Most of the limited instances of significant declines in success ratios cluster in the futures markets for gold, zinc, cocoa, corn and soybeans. In comparison, energy and livestock show no significant trending behavior. Figure 1 illustrates the success of some selected strategies applied to gold and silver futures. We see that the predictive accuracy for gold is declining significantly whereas it appears to be rather stable (with fluctuations around a mean of about 0.6) for silver. Overall, our results suggest that commodity futures' cross-sectional momentum

¹⁴ The observable distinct results for Brent and WTI crude oil are not surprising because their time-series characteristics are quite different (see Tian and Lai 2019).

¹⁵ We would not expect levels very close to one because past returns are unlikely to be able to forecast unforeseeable outside events related to climate phenomena (affecting, for example, wheat prices) or crisis-related investor behavior (influencing, for example, a safe haven asset like gold).

Table 2 Average success ratios

	Cross-sectional momentum					Time-series momentum			
	1 month	3 months	6 months	9 months	12 months	(20, 120, 0.025)	(20, 240, 0.05)	(40, 120, 0.025)	(40, 240, 0.05)
<i>Energy</i>									
Crude oil (Brent)	0.534	0.524	0.508	0.499	0.513	0.526	0.519	0.528	0.501
Crude oil (WTI)	0.544 ^b	0.555 ^a	0.565 ^a	0.539 ^c	0.567 ^a	0.513	0.486	0.512	0.488
Gas oil	0.531	0.512	0.519	0.492	0.543	0.530	0.528	0.513	0.520
Heating oil	0.546 ^b	0.531 ^c	0.538 ^c	0.534 ^c	0.548 ^b	0.509	0.502	0.489	0.481
Natural gas	0.551 ^a	0.581 ^a	0.610 ^a	0.576 ^b	0.608 ^a	0.562 ^b	0.548 ^c	0.556 ^b	0.534
Unleaded gasoline	0.492	0.488	0.483	0.478	0.520	0.492	0.509	0.495	0.521
<i>Precious metals</i>									
Gold	0.623 ^a	0.628 ^a	0.635 ^a	0.621 ^a	0.622 ^a	0.445 ^b	0.446 ^c	0.461 ^c	0.483
Silver	0.586 ^a	0.590 ^a	0.598 ^a	0.597 ^a	0.601 ^a	0.482	0.503	0.496	0.494
<i>Industrial metals</i>									
Aluminum	0.658 ^a	0.664 ^a	0.668 ^a	0.679 ^a	0.639 ^a	0.533 ^c	0.511	0.543 ^c	0.519
Copper	0.564 ^a	0.598 ^a	0.605 ^a	0.586 ^a	0.608 ^a	0.519	0.491	0.513	0.496
Lead	0.531	0.553 ^b	0.523	0.512	0.545 ^b	0.512	0.511	0.502	0.522
Nickel	0.553 ^a	0.576 ^a	0.582 ^a	0.567 ^a	0.536	0.559 ^b	0.525	0.549 ^c	0.519
Zinc	0.599 ^a	0.625 ^a	0.616 ^a	0.605 ^a	0.608 ^a	0.542 ^b	0.520	0.543 ^b	0.552 ^c

Table 2 continued

	Cross-sectional momentum					Time-series momentum			
	1 month	3 months	6 months	9 months	12 months	(20, 120, 0.025)	(20, 240, 0.05)	(40, 120, 0.025)	(40, 240, 0.05)
<i>Agriculture</i>									
Cocoa	0.537 ^b	0.583 ^a	0.594 ^a	0.591 ^a	0.598 ^a	0.475 ^c	0.458 ^c	0.477	0.505
Coffee	0.554 ^a	0.590 ^a	0.593 ^a	0.599 ^a	0.595 ^a	0.544 ^b	0.549 ^b	0.556 ^a	0.561 ^a
Corn	0.612 ^a	0.629 ^a	0.630 ^a	0.638 ^a	0.630 ^a	0.518	0.484	0.523	0.488
Cotton	0.570 ^a	0.587 ^a	0.603 ^a	0.601 ^a	0.591 ^a	0.508	0.493	0.510	0.492
Soybeans	0.622 ^a	0.594 ^a	0.612 ^a	0.609 ^a	0.614 ^a	0.458 ^b	0.479	0.470 ^c	0.487
Sugar	0.561 ^a	0.562 ^a	0.566 ^a	0.542 ^b	0.554 ^a	0.556 ^a	0.547 ^b	0.553 ^a	0.536 ^c
Wheat (Chicago)	0.625 ^a	0.621 ^a	0.604 ^a	0.612 ^a	0.625 ^a	0.516	0.498	0.510	0.513
Wheat (Kansas)	0.606 ^a	0.621 ^a	0.653 ^a	0.647 ^a	0.647 ^a	0.489	0.482	0.511	0.502
<i>Livestock</i>									
Feeder cattle	0.580 ^a	0.577 ^a	0.569 ^a	0.557 ^b	0.589 ^a	0.487	0.467	0.475	0.440 ^b
Lean hogs	0.548 ^a	0.542 ^a	0.554 ^a	0.546 ^a	0.546 ^b	0.474 ^c	0.455 ^b	0.474	0.449 ^b
Live cattle	0.586 ^a	0.571 ^a	0.568 ^a	0.565 ^a	0.562 ^a	0.526	0.503	0.519	0.508
<i>Summary</i>									
$\bar{h} < 0.5$ (sign.)	1(0)	1(0)	1(0)	3(0)	0(0)	8(4)	11(3)	8(2)	10(2)
$\bar{h} > 0.5$ (sign.)	23(20)	23(21)	23(20)	21(20)	24(20)	16(6)	13(3)	16(6)	14(3)

For our 24 commodities and different specifications of cross-sectional and time-series momentum strategies, this table reports the time-series averages of the annual success ratios defined in Sect. 3. Cross-sectional momentum is implemented for formation periods of 1, 3, 6, 9 and 12 months. Time-series momentum (or VMA) rules are specified via (s, l, w) where s and l are the length (in days) of the short-run and long-run moving averages, respectively, and w is the percentage defining the band filtering out smaller fluctuations. In all cases, the holding period is assumed to be one month. Based on typical t -tests, a , b and c imply significant differences from 0.5 at the 1%, 5% and 10% level, respectively. For an overall perspective, the bottom of the table counts the number of mean success ratios smaller/greater than 0.5 (across all commodities) and additionally gives the number of statistically significant cases in parentheses

Table 3 Constant coefficients of trend regressions

	Cross-sectional momentum					Time-series momentum			
	1 month	3 months	6 months	9 months	12 months	(20, 120, 0.025)	(20, 240, 0.05)	(40, 120, 0.025)	(40, 240, 0.05)
<i>Energy</i>									
Crude oil (Brent)	0.513	0.512	0.512	0.444	0.510	0.521	0.593	0.538	0.614
Crude oil (WTI)	0.595 ^b	0.587 ^b	0.567 ^c	0.525	0.549	0.579 ^c	0.536	0.573 ^c	0.512
Gas oil	0.529	0.524	0.530	0.486	0.508	0.538	0.572	0.538	0.577
Heating oil	0.575 ^b	0.548	0.556	0.523	0.520	0.507	0.495	0.490	0.451
Natural gas	0.557 ^c	0.537	0.552	0.480	0.542	0.518	0.487	0.496	0.460
Unleaded gasoline	0.532	0.485	0.520	0.493	0.543	0.489	0.487	0.500	0.526
<i>Precious metals</i>									
Gold	0.659 ^a	0.700 ^a	0.748 ^a	0.732 ^a	0.741 ^a	0.487	0.504	0.490	0.544
Silver	0.624 ^a	0.662 ^a	0.645 ^a	0.615 ^b	0.601 ^b	0.505	0.488	0.534	0.499
<i>Industrial metals</i>									
Aluminum	0.706 ^a	0.679 ^a	0.652 ^a	0.704 ^a	0.685 ^a	0.498	0.510	0.480	0.512
Copper	0.591 ^a	0.634 ^a	0.661 ^a	0.624 ^a	0.634 ^a	0.559	0.560	0.582 ^c	0.601 ^c
Lead	0.596 ^b	0.632 ^b	0.617 ^b	0.583	0.593 ^b	0.542	0.538	0.563	0.564
Nickel	0.547	0.575 ^b	0.597 ^b	0.608 ^b	0.615 ^b	0.585 ^c	0.591 ^c	0.587 ^c	0.573
Zinc	0.736 ^a	0.758 ^a	0.759 ^a	0.759 ^a	0.760 ^a	0.548	0.574	0.566 ^c	0.608 ^c

Table 3 continued

	Cross-sectional momentum				Time-series momentum			
	1 month	3 months	6 months	9 months	12 months	(20, 120, 0.025)	(20, 240, 0.05)	(40, 120, 0.025)
<i>Agriculture</i>								
Cocoa	0.622 ^a	0.712 ^a	0.660 ^a	0.696 ^a	0.721 ^a	0.467	0.462	0.473
Coffee	0.536	0.529	0.537	0.550	0.555	0.525	0.534	0.519
Corn	0.710 ^a	0.709 ^a	0.727 ^a	0.701 ^a	0.688 ^a	0.520	0.465	0.518
Cotton	0.590 ^a	0.602 ^b	0.630 ^a	0.627 ^b	0.625 ^a	0.545	0.499	0.523
Soybeans	0.677 ^a	0.695 ^a	0.708 ^a	0.711 ^a	0.720 ^a	0.485	0.508	0.475
Sugar	0.576 ^a	0.566 ^b	0.578 ^b	0.576 ^c	0.591 ^b	0.596 ^a	0.589 ^a	0.581 ^a
Wheat (Chicago)	0.658 ^a	0.664 ^a	0.637 ^a	0.640 ^a	0.631 ^a	0.559 ^c	0.522	0.545
Wheat (Kansas)	0.638 ^a	0.701 ^a	0.665 ^a	0.655 ^b	0.683 ^b	0.452	0.513	0.525
<i>Livestock</i>								
Feeder cattle	0.613 ^a	0.542	0.540	0.570	0.642 ^b	0.525	0.438	0.492
Lean hogs	0.519	0.564 ^b	0.529	0.549	0.548	0.614 ^a	0.601 ^b	0.596 ^a
Live cattle	0.575 ^a	0.575 ^b	0.569 ^c	0.562 ^c	0.567 ^c	0.588 ^b	0.595 ^b	0.581 ^b
<i>Summary</i>								
$\hat{\alpha} < 0.5$ (sign.)	0(0)	1(0)	0(0)	4(0)	0(0)	6(0)	8(0)	7(0)
$\hat{\alpha} > 0.5$ (sign.)	24(18)	23(17)	24(16)	20(14)	24(16)	18(6)	16(4)	16(7)

In a structure similar to Table 2 and for each of our trend-following commodity futures strategies, this table reports the constant coefficients of least squares regressions of annual success ratios on time. *a*, *b* and *c* imply significant differences from 0.5 at the 1%, 5% and 10% level, respectively. The summary at the bottom of the table gives the number of constants above/below 0.5 (across all commodities) and, in parentheses, the corresponding cases of significance

Table 4 Slope coefficients of trend regressions

	Cross-sectional momentum					Time-series momentum				
	1 month	3 months	6 months	9 months	12 months	(20, 120, 0.025)	(20, 240, 0.05)	(40, 120, 0.025)	(40, 240, 0.05)	
<i>Energy</i>										
Crude oil (Brent)	0.240	0.062	-0.078	0.526	0.188	-0.102	-0.875	-0.253	-1.263	
Crude oil (WTI)	-0.324	-0.196	-0.008	0.057	0.058	-0.397	-0.307	-0.380	-0.135	
Gas oil	0.085	-0.065	-0.121	0.067	0.429	-0.210	-0.616	-0.371	-0.740	
Heating oil	-0.147	-0.090	-0.071	0.058	0.233	0.034	0.035	0.018	0.149	
Natural gas	-0.094	0.244	0.327	0.610	0.385	0.216	0.331	0.333	0.432	
Unleaded gasoline	-0.225	0.016	-0.238	-0.109	-0.237	-0.025	0.038	-0.068	-0.130	
<i>Precious metals</i>										
Gold	-0.158	-0.327	-0.496 ^b	-0.503 ^c	-0.528 ^b	-0.218	-0.360	-0.160	-0.396	
Silver	-0.155	-0.297 ^c	-0.201	-0.079	0.005	-0.114	0.019	-0.181	-0.068	
<i>Industrial metals</i>										
Aluminum	-0.342	-0.131	0.076	-0.161	-0.177	0.200	0.130	0.395	0.177	
Copper	-0.127	-0.188	-0.282 ^c	-0.200	-0.165	-0.182	-0.326	-0.325	-0.509 ^c	
Lead	-0.558	-0.661	-0.752 ^c	-0.471	-0.431	-0.182	-0.202	-0.434	-0.311	
Nickel	0.028	-0.039	-0.120	-0.229	-0.483	-0.215	-0.337	-0.293	-0.253	
Zinc	-0.952 ^a	-0.933 ^b	-0.980 ^b	-1.023 ^a	-0.985 ^b	-0.052	-0.380	-0.181	-0.518	

Table 4 continued

	Cross-sectional momentum					Time-series momentum			
	1 month	3 months	6 months	9 months	12 months	(20, 120, 0.025)	(20, 240, 0.05)	(40, 120, 0.025)	(40, 240, 0.05)
Agriculture									
Cocoa	-0.461 ^b	-0.683 ^a	-0.338	-0.580 ^c	-0.638 ^b	0.073	-0.025	0.052	-0.253
Coffee	0.101	0.326 ^c	0.301	0.245	0.269	0.076	0.022	0.172	0.125
Corn	-0.353 ^a	-0.349 ^b	-0.400 ^b	-0.203	-0.178	-0.027	0.044	-0.016	0.050
Cotton	-0.112	-0.094	-0.148	-0.136	-0.111	-0.184	0.031	-0.071	0.045
Soybeans	-0.248 ^b	-0.356 ^b	-0.417 ^b	-0.438 ^b	-0.453 ^a	-0.108	-0.145	-0.042	-0.071
Sugar	-0.067	-0.020	-0.037	-0.115	-0.106	-0.180	-0.222	-0.135	-0.201
Wheat (Chicago)	-0.163	-0.202	-0.099	-0.141	-0.057	-0.183	-0.099	-0.142	-0.009
Wheat (Kansas)	-0.400	-0.873	-0.223	-0.205	-0.360	0.320	-0.065	-0.138	-0.153
Livestock									
Feeder cattle	-0.482	0.314	0.181	-0.317	-0.857	-0.432	0.218	-0.136	-0.221
Lean hogs	0.124	-0.092	0.118	0.002	0.001	-0.630 ^a	-0.613 ^a	-0.545 ^a	-0.538 ^a
Live cattle	0.078	0.029	0.041	0.060	0.027	-0.206	-0.401 ^b	-0.275 ^c	-0.376 ^c
Summary									
$\hat{\beta} < 0$ (sign.)	18(4)	18(5)	18(6)	16(4)	15(4)	18(1)	15(2)	19(2)	18(3)
$\hat{\beta} > 0$ (sign.)	6(0)	6(1)	6(0)	8(0)	9(0)	6(0)	9(0)	5(0)	6(0)

Extending Table 3, this table reports the slope coefficients of least squares regressions of annual success ratios on time. For better visualization, the coefficient estimates have been multiplied by 100. *a*, *b* and *c* imply significance at the 1%, 5% and 10% level, respectively. The summary at the bottom of the table gives the number of negative/positive trends (across all commodities) and, in parentheses, the corresponding cases of significance

is likely to remain as effective as documented in the past, which contrasts with the recent evidence in stock markets (see Chordia et al. 2014).¹⁶

Turning to the time-series momentum strategies (and focusing on high significance of 1% and 5% only), we document high initial forecasting performance only for sugar, lean hogs and live cattle. Similar to cross-sectional momentum, significant trending behavior is widely absent. The few cases of significant decline cluster in the livestock sector. Because the corresponding average success ratios in Table 2 are below 0.5, these findings call for caution in a future use of time-series momentum strategies for livestock contracts. For these contracts, the strategies may have had suitable predictive power in the past, but apparently it has started to vanish. Figure 1, which covers success ratios and corresponding trend lines for selected time-series momentum strategies for gold and silver, is representative for most other sectors. While stock market success ratios exhibit declines from initially high levels (see Strobel and Auer 2018), in futures markets, the strategies are less persuasive with respect to forecasting ability and show no significant trending behavior. This result has to be interpreted carefully because it does not mean that the strategies are useless. Their directional accuracy may be similar to tossing a coin but, given the results of the previous literature, this interestingly appears to be sufficient to provide beneficial investment performance when applied to the majority of available commodity futures (see Szakmary et al. 2010; Clare et al. 2014). The observation of not existing trends in success ratios is important because it conveys a sense of stability and thus continuous predictive reliability.

4.2 Robustness checks

Because some features of our research design might be thought of driving its main outcome, we perform several sensitivity checks to ensure robustness. Because these checks largely confirm the findings of the previous Section, we concentrate on a verbal summary of their basic idea.¹⁷

Rollover Miffre and Rallis (2007) analyze whether rollover modifications (such as a later rollover date and using more distant maturity contracts) have a damaging impact on momentum rules and find that, even though profits are lower, momentum persists. Nevertheless, one might argue that the type of rollover implemented in GSCI indices may influence our trend results. To rule this out, we collect the individual futures data of the indices and manually implement alternative rollovers. Specifically, we follow Lukac et al. (1988) by rolling over into the next contract month prior to delivery and, from this point on, using data of the new contract for signal generation. We also follow Fuertes et al. (2015) by immediately rolling over at the last trading day. However, we find that our results are not sensitive to such modifications.

Cross-sectional momentum While it has become standard to rank assets based on cumulative past returns, some studies use alternative variables. For example, Rachev et al. (2007) employ reward-to-risk measures (like the Sharpe ratio) and Narayan

¹⁶ For a detailed empirical analysis of the reasons why momentum strategies behave differently in stock and futures markets, see Chevallier et al. (2013).

¹⁷ Detailed results are available from the authors upon request.

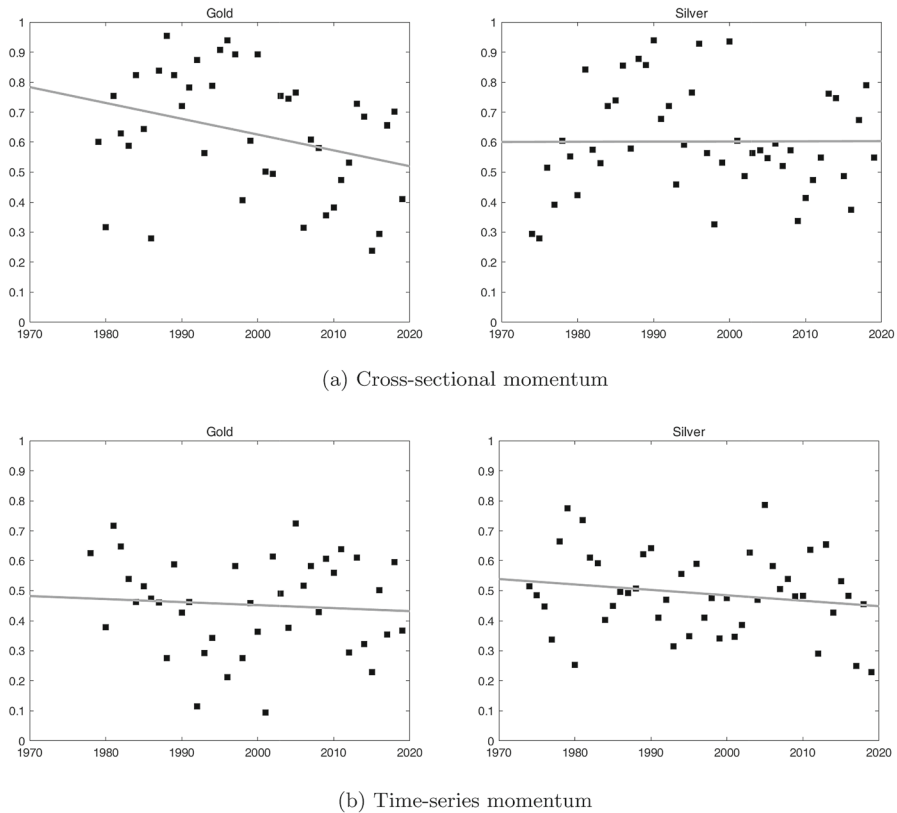


Fig. 1 Exemplary success ratio trend plots. For gold and silver futures, this figure plots the annual success ratios of selected trend-following trading rules and the corresponding least squares trend regression lines. **a** relates to cross-sectional momentum with a 12-month formation period. **b** captures the time-series momentum (or VMA) rule ($s = 40$, $l = 120$, $w = 0.025$). The holding period is one month

et al. (2015) implement the momentum indicator (the difference between short-run and long-run moving return averages). Furthermore, while we followed Szakmary et al. (2010) by dividing assets into three groups (top, middle, bottom) and using only the top and bottom thirds, Miffre and Fernandez-Perez (2015) suggest concentrating on the top and bottom fifths.¹⁸ In contrast to this focus on relative strength, where a future with negative past performance may show up on the winner side because it is less negative than others, Erb and Harvey (2006) use positive and negative returns to draw a clearer distinction between winners and losers. Overall, such settings do not alter our conclusions.

Time-series momentum Occasionally, moving average rules are implemented by calculating the moving average of returns instead of prices (see Ratner and Leal 1999).

¹⁸ Adding more futures to the winner or loser sides enhances risk diversification at the cost of lowering the dispersion of returns between the best and worst performing futures and thus the profitability of the strategy.

Furthermore, one might also use exponentially weighted averages instead of arithmetic averages (see Rosillo et al. 2013).¹⁹ Applying these alternative approaches reveals that return-based moving averages lead to lower numbers of successful predictions and that our results remain largely unchanged when switching to exponential moving averages.

Signal frequency and holding period Because our analysis provides signals on a daily frequency, we have also modified our setup to generate only one signal per month at the end of the month (as in Szakmary et al. 2010) while leaving all other settings untouched. Furthermore, we repeated our calculations for daily and weekly holding period lengths. Again, our main conclusions hold but with somewhat fewer instances of statistically significant trends.

Trend function Finally, we have estimated an exponential and a quadratic trend model in addition to our standard setting. In the former, we use the natural logarithm of the success ratio as the left-hand variable; in the latter, we add time squared as a second right-hand variable (see Pindyck and Rubinfeld 1998). This way, we can consider a potentially nonlinear evolution, but find that it does not change the signs and significances of our trend results.

5 Conclusion

Past research has documented impressive performance of trend-following trading rules in both the stock and the futures markets. However, more recent studies indicate that the benefits of such strategies in stock markets tends to decline with falling arbitrage barriers (lower transaction costs, market access for low-sum investors) and increased (automated) trading activity over time (see Strobel and Auer 2018; Auer and Rottmann 2019). Motivated by such results, we have analyzed whether similar tendencies can be observed for commodity futures markets.

Using a broad selection of energy, precious metals, industrial metals, agriculture and livestock futures, we have tested for significant time trends in the success ratios of various cross-sectional and time-series momentum strategies. In other words, we have revealed the predictive capabilities behind the financial performance excessively documented in the previous research and investigated their dynamics over time. Our findings for cross-sectional momentum indicate predictive accuracy above the level of a simple coin toss and only few instances of trending. In contrast, the success ratios of time-series momentum are close to tossing a fair coin but show a similar absence of significant time trends. Supported by a variety of robustness checks, these results have two important implications for commodity investing. First, time-series momentum increases investors' wealth despite having success ratios similar to a fair coin. Second, contrasting stock market research, historically beneficial trend-following strategies can be expected to continue to perform well in the future (unless, of course, there are drastic changes in market structure or investor behavior).

¹⁹ Adaptive averages have also received attention. They seek to identify and adopt changing market conditions via an efficiency ratio derived from the notion of fractal efficiency and a method close to rescaled range analysis (see Ellis and Parbery 2005). However, the corresponding new trading rules leave the classic VMA framework.

As far as future research is concerned, several issues merit deeper investigation. First, since we have downward trends in stock markets but (almost) no trends in commodity futures markets, it may be instructive to see how this affects the predictive abilities of strategies applying momentum across asset classes (see, for example, Georgopoulou and Wang 2017). Second, because we have concentrated only on one possibility of implementing time-series momentum, our study may be extended to regression-based forms (as in, for example, Moskowitz et al. 2012). Finally, we might wish to reveal the origins of the significant trends detected for some commodities (such as gold) by looking at investor sentiment variables (see Jacobs 2015) or the impact of financialization on momentum performance (see Zaremba 2016).

Acknowledgements The author thanks Horst Rottmann, Julia Mehlitz, Anja Vinzelberg and an anonymous reviewer for valuable comments and suggestions. He is also indebted to the Fritz Thyssen Stiftung (Grant 20.18.0.016WW) for generous financial support.

References

- Auer, B.: Does the choice of performance measure influence the evaluation of commodity investments? *Int. Rev. Financ. Anal.* **38**, 142–150 (2015)
- Auer, B., Rottmann, H.: Have capital market anomalies worldwide attenuated in the recent era of high liquidity and trading activity? *J. Econ. Bus.* **103**, 61–79 (2019)
- Bajgrowicz, P., Scaillet, O.: Technical trading revisited: false discoveries, persistence tests and transaction costs. *J. Financ. Econ.* **106**(3), 473–491 (2012)
- Bianchi, R., Drew, M., Fan, J.: Combining momentum with reversal in commodity futures. *J. Bank. Finance* **59**, 423–444 (2015a)
- Bianchi, R., Drew, M., Fan, J.: Microscopic momentum in commodity futures. Griffith University Discussion Papers Finance No. 2015-10 (2015b)
- Bianchi, R., Drew, M., Fan, J.: Commodities momentum: a behavioral perspective. *J. Bank. Finance* **72**, 133–150 (2016)
- Brock, W., Lakonishok, J., LeBaron, B.: Simple technical trading rules and the stochastic properties of stock returns. *J. Finance* **47**(5), 1731–1764 (1992)
- Chevallier, J., Gatumel, M., Ielpo, F.: Understanding momentum in commodity markets. *Appl. Econ. Lett.* **20**(15), 1383–1402 (2013)
- Chordia, T., Subrahmanyam, A., Tong, Q.: Have capital market anomalies attenuated in the recent era of high liquidity and trading activity? *J. Account. Econ.* **58**(1), 41–58 (2014)
- Clare, A., Seaton, J., Smith, P., Thomas, S.: Trend following, risk parity and momentum in commodity futures. *Int. Rev. Financ. Anal.* **31**, 1–12 (2014)
- de Groot, W., Karstanje, D., Zhou, W.: Exploiting commodity momentum along the futures curves. *J. Bank. Finance* **48**, 79–93 (2014)
- Ellis, C., Parbery, S.: Is smarter better? A comparison of adaptive, and simple moving average trading strategies. *Res. Int. Bus. Finance* **19**(3), 399–411 (2005)
- Erb, C., Harvey, C.: The strategic and tactical value of commodity futures. *Financ. Anal. J.* **62**(2), 69–97 (2006)
- Fama, E., French, K.: Dissecting anomalies. *J. Finance* **63**(4), 1653–1678 (2008)
- Fama, E., French, K.: Size, value, and momentum in international stock returns. *J. Financ. Econ.* **105**(3), 457–472 (2012)
- Fifield, S., Power, D., Knipe, D.: The performance of moving average rules in emerging stock markets. *Appl. Financ. Econ.* **18**(19), 1515–1532 (2008)
- Fleming, J., Ostdiek, B., Whaley, R.: Trading costs and the relative rates of price discovery in stock, futures, and option markets. *J. Futures Mark.* **16**(4), 353–387 (1996)
- Foltice, B., Langer, T.: Profitable momentum trading strategies for individual investors. *Financ. Mark. Portf. Manag.* **29**, 85–113 (2015)

- Fong, W., Wong, W., Lean, H.: International momentum strategies: a stochastic dominance approach. *J. Financ. Mark.* **8**(1), 89–109 (2005)
- Fuertes, A., Miffre, J., Fernandez-Perez, A.: Commodity strategies based on momentum, term structure and idiosyncratic volatility. *J. Futures Mark.* **35**(3), 274–295 (2015)
- Fuertes, A., Miffre, J., Rallis, G.: Tactical allocation in commodity futures markets: combining momentum and term structure signals. *J. Bank. Finance* **34**(10), 2530–2548 (2010)
- Georgopoulou, A., Wang, J.: The trend is your friend: time-series momentum strategies across equity and commodity markets. *Rev. Finance* **21**(4), 1557–1592 (2017)
- Gorton, G., Rouwenhorst, K.: Facts and fantasies about commodity futures. *Financ. Anal. J.* **62**(2), 47–68 (2006)
- Hendershott, T., Jones, C., Menkveld, A.: Does algorithmic trading improve liquidity? *J. Finance* **66**(1), 1–33 (2011)
- Hong, H., Yogo, M.: What does futures market interest tell us about the macroeconomy and asset prices? *J. Financ. Econ.* **105**(3), 473–490 (2012)
- Irwin, S., Yoshimaru, S.: Managed futures, positive feedback trading, and futures price volatility. *J. Futures Mark.* **19**(7), 759–776 (1999)
- Jacobs, H.: What explains the dynamics of 100 anomalies? *J. Bank. Finance* **57**, 65–85 (2015)
- Jacobs, H., Müller, S.: Anomalies across the globe: once public, no longer existent? *J. Financ. Econ.* **135**(1), 213–230 (2020)
- Jegadeesh, N., Titman, S.: Returns to buying winners and selling losers: implications for stock market efficiency. *J. Finance* **48**(1), 65–91 (1993)
- Jegadeesh, N., Titman, S.: Profitability of momentum strategies: an evaluation of alternative explanations. *J. Finance* **56**(2), 699–720 (2001)
- Katusime, L., Shamsuddin, A., Agbola, F.: Foreign exchange market efficiency and profitability of trading rules: evidence from a developing country. *Int. Rev. Econ. Finance* **35**, 315–332 (2015)
- Kavajecz, K., Odders-White, E.: Technical analysis and liquidity provision. *Rev. Financ. Stud.* **17**(4), 1043–1071 (2004)
- Kho, B.: Time-varying risk premia, volatility, and technical trading rule profits: evidence from foreign currency futures markets. *J. Financ. Econ.* **41**(2), 249–290 (1996)
- Korajczyk, R., Sadka, R.: Are momentum profits robust to trading costs? *J. Finance* **59**(3), 1039–1082 (2004)
- Lesmond, D., Schill, M., Zhou, C.: The illusory nature of momentum profits. *J. Financ. Econ.* **71**(2), 349–380 (2004)
- Locke, P., Venkatesch, P.: Futures market transaction costs. *J. Futures Mark.* **17**(2), 229–245 (1997)
- Lubnau, T., Todorova, N.: Trading on mean-reversion in energy futures markets. *Energy Econ.* **51**, 312–319 (2015)
- Lukac, L., Brorsen, B., Irwin, S.: A test of futures market disequilibrium using twelve different technical trading systems. *Appl. Econ.* **20**(5), 623–639 (1988)
- Marshall, B., Cahan, R., Cahan, J.: Can commodity futures be profitably traded with quantitative market timing strategies? *J. Bank. Finance* **32**(9), 1810–1819 (2008)
- McLean, R., Pontiff, J.: Does academic research destroy stock return predictability? *J. Finance* **71**(1), 5–32 (2016)
- Miffre, J.: Long-short commodity investing: a review of the literature. *J. Commod. Mark.* **1**(1), 3–13 (2016)
- Miffre, J., Fernandez-Perez, A.: The case for long-short commodity investing. *J. Altern. Invest.* **18**(1), 92–104 (2015)
- Miffre, J., Rallis, G.: Momentum strategies in commodity futures markets. *J. Bank. Finance* **31**(6), 1863–1886 (2007)
- Morana, C.: A semiparametric approach to short-term oil price forecasting. *Energy Econ.* **23**(3), 325–338 (2001)
- Moskowitz, T., Ooi, Y., Pedersen, L.: Time series momentum. *J. Financ. Econ.* **104**(2), 228–250 (2012)
- Narayan, P., Ahmed, H., Narayan, S.: Do momentum-based trading strategies work in the commodity futures markets? *J. Futures Mark.* **35**(9), 868–891 (2015)
- Narayan, P., Narayan, S., Sharma, S.: An analysis of commodity markets: what gain for investors? *J. Bank. Finance* **37**(10), 3878–3889 (2013)
- Olson, D.: Have trading rule profits in the currency markets declined over time? *J. Bank. Finance* **28**(1), 85–105 (2004)

- Park, C., Irwin, S.: The profitability of technical trading rules in US futures markets: a data snooping free test. AgMAS Project Research Report 2005-04 (2005)
- Park, C., Irwin, S.: A reality check on technical trading rule profits in the U.S. futures markets. *J. Futures Mark.* **30**(7), 633–659 (2010)
- Park, C., Irwin, S.: What do we know about the profitability of technical analysis? *J. Econ. Surv.* **21**(4), 786–826 (2007)
- Pindyck, R., Rubinfeld, D.: *Econometric Models and Economic Forecasts*. McGraw-Hill, Singapore (1998)
- Rachev, S., Jašić, T., Stoyanov, S., Fabozzi, F.: Momentum strategies based on reward-risk stock selection criteria. *J. Bank. Finance* **31**(8), 2325–2346 (2007)
- Ratner, M., Leal, R.: Tests of technical trading strategies in the emerging equity markets of Latin America and Asia. *J. Bank. Finance* **23**(12), 1887–1905 (1999)
- Rosillo, R., de la Fuente, D., Brugos, J.: Technical analysis and the Spanish stock exchange: testing the RSI, MACD, momentum and stochastic rules using Spanish market companies. *Appl. Econ.* **45**(12), 1541–1550 (2013)
- Shen, Q., Szakmary, A., Sharma, S.: An examination of momentum strategies in commodity futures markets. *J. Futures Mark.* **27**(3), 227–256 (2007)
- S&P Dow Jones Indices: S&P GSCI Methodology. McGraw Hill Financial, New York (2016)
- Strobel, M., Auer, B.: Does the predictive power of variable moving average rules vanish over time and can we explain such tendencies? *Int. Rev. Econ. Finance* **53**, 168–184 (2018)
- Swinkels, L.: Momentum investing: a survey. *J. Asset Manag.* **5**, 120–143 (2004)
- Szakmary, A., Shen, Q., Sharma, S.: Trend-following trading strategies in commodity futures: a re-examination. *J. Bank. Finance* **34**(2), 409–426 (2010)
- Taylor, N.: The rise and fall of technical trading rule success. *J. Bank. Finance* **40**, 286–302 (2014)
- Taylor, S.: Stock index and price dynamics in the UK and the US: new evidence from a trading rule and statistical analysis. *Eu. J. Finance* **6**(1), 39–69 (2000)
- Tian, H., Lai, W.: The causes of stage expansion of WTI/Brent spread. *Pet. Sci.* **16**, 1493–1505 (2019)
- Wang, C., Yu, M.: Trading activity and price reversals in futures markets. *J. Bank. Finance* **28**(6), 1337–1361 (2004)
- Wang, S., Yu, L., Lai, K.: A novel hybrid AI system framework for crude oil price forecasting. In: Shi, Y., Xu, W., Chen, Z. (eds.) *Data Min. Knowl. Manag.*, pp. 233–242. Springer, Berlin, Heidelberg (2004)
- Yao, Y.: Momentum, contrarian, and the January seasonality. *J. Bank. Finance* **36**(10), 2757–2769 (2012)
- Zaremba, A.: Strategies based on momentum and term structure in financialized commodity markets. *Bus. Econ. Res. J.* **7**(1), 31–46 (2016)
- Zhang, H., Auer, B., Vortelinos, D.: Performance ranking (dis)similarities in commodity markets. *Glob. Finance J.* **35**, 115–137 (2018)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Benjamin R. Auer is Full Professor of Finance at the Brandenburg University of Technology Cottbus-Senftenberg, lecturer for financial data management at the University of Leipzig and Research Affiliate of the CESifo Group Munich. He has published in top-tier journals such as *Management Science*, the *European Journal of Operational Research* and the *Review of Derivatives Research* and has recently been ranked place 9 among all German-speaking researchers below the age of 40 in the *Handelsblatt/Wirtschaftswoche Research Ranking 2020*. His main areas of expertise are investment management, risk and performance measurement as well as applied financial econometrics.